

PAUL KARONJI WAITHAKA

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-stack engineer shipping production systems with live users: a hospital management platform running in 3 hospitals, SaaS products processing real card payments, and an offline-first POS suite with KRA eTIMS fiscalization. Final-year BSc Software Development student (KCA University, expected July 2027) with 3+ years building across Kenya's payment rails (M-Pesa Daraja, Paystack), offline-first architecture, real-time systems, and AI agent tooling (MCP). Works fully remote and async — all client and team engagements to date.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Go, PHP, Kotlin, Python, SQL, Bash

Backend: Node.js, Express, NestJS, Laravel 10/12, Go (Chi, pgx/pgxpool, Gorilla WebSocket), REST APIs, WebSockets/Socket.IO, Prisma, Eloquent ORM

Frontend: React 18/19, Next.js 14–16, React Native (Expo), Electron, Tailwind CSS, shadcn/ui, TanStack Query, PWA, Leaflet

Databases & Cache: PostgreSQL, MySQL, SQLite, TiDB Serverless, Supabase, Firebase, Upstash Redis, Pinecone

Cloud & DevOps: Vercel, Docker, GitHub Actions CI/CD, Cloudflare R2, Render, Oracle Cloud, Neon, Caddy, Vercel Cron

Payments & Kenya Rails: M-Pesa Daraja (STK Push, C2B callbacks), Paystack (card), KRA eTIMS (OSCU/VSCU fiscalization), Africa's Talking SMS, WhatsApp Cloud API

AI & Integrations: Multi-model LLM orchestration (AI SDK: Anthropic, OpenAI, Groq, Mistral, OpenRouter), MCP server development, Google Calendar API v3, Cloudfinary, Resend, VAPID Push

Security: JWT, Row-Level Security, CSRF, CSP, rate limiting, webhook signature verification (raw-byte), HMAC-SHA256 timing-safe flows, hardware fingerprinting, CVE patching, Zod

Architecture: Offline-first, multi-tenant SaaS, white-labeled instance deployment, hardware licensing, real-time event streaming, monorepo (Turborepo)

Testing: Jest, Vitest, Playwright, Supertest, Go testing, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Lead Engineer — WIK Technologies

Nairobi, Kenya | Oct 2025 – Present

Built and lead a software company whose products serve live users today: a hospital management platform running in 3 hospitals and a production POS platform — working with a small engineering team.

- Architected WIK-POS, an offline-first desktop POS suite (Electron + React + TypeScript + SQLite): hardware-bound JWT licensing (CPU + MAC fingerprint), M-Pesa Daraja STK Push with IP-guarded callbacks, and KRA eTIMS fiscalization (OSCU worker, AES-256-encrypted tax credentials) — modular editions for retail, hotel F&B (KOT), and pharmacy (PPB batch tracking, 60-day FEFO quarantine); production-deployed at pos.wiktechnologies.com.
- Built Pos-License-Server, a centralized licensing authority with a 3-tier model (lifetime / subscription / trial), hardware fingerprinting, and remote revocation — Docker + Caddy deployment across Render/Oracle Cloud with Neon PostgreSQL and automated Cloudflare R2 backups.
- Led engineering of WIK HMS, a hospital management system (Laravel 12, React 19) live in 3 hospitals via white-labeled per-client instances: 200+ REST endpoints, 56 migrations, Observer-pattern automated billing, real-time triage queues, and FEFO pharmacy registers; live demo at hmisdemo.wiktechnologies.com.

Founding Engineer — Atlas Data (FleetTrack)

Remote | Aug 2026 – Present

- Sole founding engineer of a multi-tenant fleet-telematics and dispatch platform, built end-to-end: high-concurrency Go backend (Chi, pgxpool, Gorilla WebSocket), React 19 web console, and Expo/React Native driver app; deployed in active testing at fleetops-mauve.vercel.app.
- Built the offline-first telemetry pipeline: driver devices buffer GPS/battery samples in SQLite and sync in batches on reconnect; a PostgreSQL hot-state cache feeds real-time WebSocket fan-out to dispatchers' live maps.
- Shipped automated transit-safety systems: 30-minute stoppage detection auto-raises critical incidents and live-map alerts; QR/passcode dispatch claiming; pre/post-trip inspections that lock vehicles on critical failures — with 4-role RBAC and sequential SQL migrations.

Technical Lead (Industrial Attachment — internship) — JHUB Africa

Nairobi, Kenya | Sep 2025 – Dec 2025

- Led a development team building the JHUB Africa Innovation Tracker (innovation.jhubafrica.com), a PHP/MySQL/Docker platform for JKUAT Hub: 4 user roles, 6-stage project lifecycle, mentor-assignment engine, document management, and activity audit logs — delivered on schedule and live in active use.

- Established team coding standards (PSR-12) and ran code reviews.

Freelance Software Developer

Nairobi, Kenya (Remote) | 2023 – Present

- Delivered 10+ end-to-end products for clients — including K-Track (tutor-agency SaaS, below), a flower-shop commerce system with M-Pesa STK Push integration, tenant Wi-Fi billing (Hotbill), a school ERP (Wikscholar), and a logistics/waybill system (Logit) — owning requirements, architecture, deployment, and maintenance.
- Ongoing maintenance relationships with several clients; zero missed deadlines across engagements.

SELECTED PROJECTS

K-Track — Tutor-Agency & Commission SaaS

live: ktrack.vercel.app — freelance, Jun 2025

React 19, Node.js/Express 5, TypeScript, MySQL/TiDB Serverless, Socket.IO, Cloudflare R2, Paystack

- Live SaaS for commission-based tutor agencies processing real Visa card payments via Paystack with USD→KES conversion; 94% payment collection rate across client billing.
- Security-hardened payments: HMAC-SHA256 timing-safe guest checkout URLs (crypto.timingSafeEqual), Paystack webhook verification against raw bytes, HttpOnly refresh cookies, scoped file authorization.
- Idempotent boot-time schema-patch service for TiDB Serverless; task pooling/claiming and real-time Socket.IO chat across superadmin/tutor/client roles.

DueSync — Task-Management PWA & MCP AI Server

live: duesync.vercel.app — open source

Next.js 15, TypeScript, Prisma, PostgreSQL, NextAuth v5, Upstash Redis, Vercel Cron

- Ships a native Model Context Protocol server (stdio + Streamable HTTP) exposing 13 tools so AI agents can create, complete, and auto-schedule deep-work tasks.
- Two-way Google Calendar sync (OAuth 2.0, proactive token refresh), recurring tasks, Pomodoro mode, timezone-aware reminders, VAPID push; PWA + Android TWA.
- Hardened: RLS on all tables, strict CSP, Upstash Redis rate limiting, Zod validation; proactively patched CVE-2025-55182 / CVE-2025-66478 (Dec 2025 React/Next.js RSC RCE advisory).

Hiro — Self-Hosted Multi-Model AI Agent

open source

Node.js/TypeScript, AI SDK, PostgreSQL/SQLite, Pinecone, WhatsApp, Telegram

- Multi-model agent mesh routing across Anthropic/OpenAI/Groq/Mistral/OpenRouter with parallel task execution and a dynamic self-improving skills system.
- Document ingestion (PDF/DOCX), web search, memory-backed conversations, voice I/O, and rich-output browser canvas across dual Telegram/WhatsApp interfaces.

MtaaKeys — Property Marketplace & Rental Management

live: mtaakeys.com

Next.js, Express 5, TypeScript, PostgreSQL/Prisma, Socket.IO, Paystack, Africa's Talking

- Marketplace + rental management with 5 user roles and row-level security; dual Paystack systems (tiered agent subscriptions + pay-per-use listing Boosts) with node-cron lifecycle jobs for renewals and downgrades.
- Algorithmic featured-listing curation (view/inquiry scoring, recency decay); real-time Socket.IO messaging; Africa's Talking SMS + Resend email notifications.

wiktrack — M-Pesa Income Tracker (Android)

- Expo/React Native app with a custom Kotlin module parsing incoming M-Pesa SMS into structured ledgers; local-first SQLite with background Supabase sync and FCM push.

Also: biashara-hub (Medusa.js v2 headless commerce with Daraja integration) • Votewatch (NestJS election-integrity platform, in active development)

EDUCATION

BSc Software Development — KCA University

Nairobi, Kenya | Sep 2021 – Expected July 2027

CERTIFICATIONS

- Cybersecurity Fundamentals — IBM SkillsBuild (2025)
- Data Protection Basics — Atingi eLearning (2025)
- Zoho Creator Developer Certification — Zoho (2025)